



PES UNIVERSITY

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Capstone Project Report Phase - 2

on

PESU - Exam Duty Allocator

Submitted by

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March 2025 - June 2025

Under the guidance of

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**FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER APPLICATIONS
PROGRAM - MASTER OF COMPUTER APPLICATIONS**

Certificate

This is to certify that the project entitled

PESU - Exam Duty Allocator

is a bonafide work carried out by

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in partial fulfillment for the completion of **Capstone Project Phase - 2** work in the Program of Study MCA under rules and regulations of **PES University**, Bengaluru during the period **March 2025 - June 2025**. The project report has been approved as it satisfies the academic requirements of 4th semester MCA.

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Declaration

I, **Manoj S**, bearing **PES1PG23CA078** hereby declare that the project entitled, **PESU - Exam Duty Allocator**, is an original work done by me under the guidance of **Mr. Dilip Kumar Maripuri**, Associate Professor, PES University and is being submitted in partial fulfillment of the requirements for completion of 4th Semester course work in the Program of Study MCA.

All corrections/suggestions indicated for internal assessment have been incorporated in the report.

The plagiarism check has been done for the report and is below the given threshold. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

Place: Bengaluru

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Abstract

It can be time-consuming and prone to mistakes for institutions to manually allocate faculty responsibilities and schedule exam times. Inefficiencies, scheduling problems, and needless administrative burdens are frequently the results. We created a web-based system that automates the process of creating exam schedules and assigning faculty responsibilities in order to address these problems.

The project's main objective is to expedite the scheduling procedure as a whole while maintaining equity, openness, and efficient resource utilization. In order to create exam schedules that are free of conflicts and divide invigilation responsibilities fairly, the system considers a number of limitations, such as faculty availability, subject competence, and institutional policies.

The backend was developed using Flask (Python) for managing core logic and API services, and React.js as the frontend for offering a responsive and seamless user interface. MongoDB was utilized for efficient data storage and management. A customized scheduling algorithm was created to offer fair distribution of faculty workload and prevent scheduling conflicts during the examination.

The system's primary features include a duty swap portal that enables professors to seek assignment adjustments if necessary and automated notifications via email API. The approach ensured a balanced allocation of tasks and greatly decreased scheduling conflicts during testing. The outcomes demonstrated increased exam preparation clarity, decreased manual labor, and better resource utilization. All things considered, it turned out to be a dependable instrument for better handling assignments and tests.

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PESU - EXAM DUTY ALLOCATOR

Chapter 1: Introduction

Chapter 1

Introduction

1.1 Problem Scenario

Manually managing faculty duty assignments and exam schedules is ineffective, time-consuming, and prone to mistakes. The examination process is disrupted by scheduling disputes, unequal workload distribution, and **last-minute alterations generate administrative burdens**. Current solutions necessitate a great deal of manual involvement since they lack automation and real-time updates. By streamlining scheduling, guaranteeing equitable duty distribution, and cutting down on administrative burden, an automated test Timetable and Duty Allocator system tackles these issues and eventually improves test management's effectiveness and transparency.

1.2 Proposed Solution

In order to avoid conflicts and guarantee equitable distribution, the suggested Exam Timetable and Duty Allocation System uses rule-based algorithms to automate **exam scheduling** and **faculty duty assignments**. Constructed using Python Flask for the backend and React.js for the frontend, it incorporates automatic notifications and real-time validation to expedite the procedure. In contrast to manual approaches, this solution **improves exam management efficiency**, **decreases administrative workload**, and reduces errors.

1.3 Purpose

By automating the entire test schedule generation and task allocation process, this initiative aims to provide conflict-free scheduling and equitable, balanced workload distribution among faculty members. The initiative greatly improves the accuracy, transparency, and general effectiveness of exam management in educational institutions by utilizing cutting-edge technology and system

intelligence. Through automated logic, real-time validation, and organized workflows, this system seeks to remove human mistake, time-consuming coordination, and communication gaps that are frequently present in the manual exam scheduling and task assignment process. By providing features like duty swap requests, real-time notifications, and role-based access to expedite departmental activities, the system also lessens the administrative burden.

1.4 Scope

By automating the creation of exam schedules and the methodical distribution of faculty duties, this initiative guarantees conflict-free scheduling and fair workload distribution. It is especially designed for educational establishments where precise planning and prompt implementation are crucial. Advanced features like **duty swap management**, real-time **notifications**, and continuous **validation** are integrated into the system to minimize delays and lower the possibility of scheduling conflicts or administrative errors. In order to ensure accountability and transparency, faculty members can request duty swaps, which are then verified by a multi-level approval procedure that includes peers, department coordinators, and the Controller of Examinations (COE).

PESU - EXAM DUTY ALLOCATOR

Chapter 2: Literature Survey

Chapter 2

Literature Survey

2.1 Domain Survey

2.1.1 Introduction to the Domain

With a particular emphasis on automated scheduling systems, this project falls under the broad category of educational technology, or EdTech. Exam scheduling and faculty management are important administrative responsibilities at educational institutions. Spreadsheets and paper-based systems have historically been used to manage these procedures manually, which is laborious and prone to human mistake. Problems including conflicting schedules, unequal workload distribution among professors, and overlapping responsibilities frequently occur, underscoring the need for a more dependable and effective strategy.

By utilizing algorithmic planning and rule-based logic, automated scheduling systems offer an answer to these problems. These systems can manage a variety of limitations, including subject matter expertise, room capacity, and faculty availability, while guaranteeing equitable and ideal task distribution. Automation greatly lessens the administrative burden and the likelihood of scheduling mistakes. Because faculty members may be guaranteed a fair distribution of responsibilities, it also increases transparency and trust in the scheduling process.

In order to further improve scheduling accuracy, contemporary scheduling solutions are progressively integrating optimization techniques, constraint fulfillment algorithms, and artificial intelligence (AI) as a result of technological advancements. These systems are able to adjust dynamically to changes, such as requests for faculty leave or last-minute replacements, guaranteeing uninterrupted academic operations. Administrators and employees can also easily utilize and access these tools thanks to web-based interfaces and real-time dashboards.

2.1.2 Core Terminologies and Definitions

- **Automated Scheduling** – Minimizing manual intervention by allocating work, resources, or events in an optimal way through the use of algorithms.
- **Faculty Duty Allocation** – Faculty members being given supervisory or invigilation responsibilities during exams.
- **Constraint-Based Scheduling** – A scheduling strategy that guarantees all limitations—like workload and availability—are fulfilled.
- **Web-Based System** – An application that can be accessed and managed remotely through a web browser.
- **Notification System** – A function that automatically notifies faculty members via email or SMS of changes to their schedules and duty assignments.

2.1.3 Associated Theoretical Concepts

- **Constraint Satisfaction Problem** – Makes certain that scheduling restrictions like room capacity, workload distribution, and teacher availability are fulfilled.
- **Greedy and Heuristic Algorithms** – Used to efficiently distribute tests and faculty responsibilities in order to optimize timetable generation.
- **Load Balancing** – Prevents workload inequalities by equitably allocating invigilation responsibilities among teachers.

2.1.4 Challenges and Open Problems

- **Dynamic Constraints Handling** – Real-time flexibility is necessary to deal with unforeseen circumstances like teacher unavailability or exam postponement.
- **Fairness in Duty Allocation** – Distributing exam responsibilities fairly while taking workload limitations and instructor preferences into account.

2.2 Literature Survey

2.2.1 Research Review

1. Anith Kumar Vadlamudi, Sushma Thigulla, Meghana Veeramachaneni, Shiva krishna Chaitanya Reddy Vangimalli, Bhaskara Rao Pondugala, "EXAM INVIGILATION SCHEDULER" 2024 International Conference on Circuit Power and Computing Technologies (ICCPCT).

Available: ResearchGate.

- **Abstracted Summary**

By automating the distribution of invigilation tasks, the Exam Invigilation Scheduler solves the inefficiencies associated with human scheduling. Manual approaches take a lot of time, are prone to mistakes, and frequently lead to an unequal workload distribution among teachers. By taking into account important variables like faculty availability, exam schedules, and workload balance, this method guarantees equity, effectiveness, and flexibility while eventually simplifying the scheduling process for educational institutions.

- **Problem Statement**

Manual invigilation scheduling is inefficient, error-prone, and challenging to manage for large institutions, frequently resulting in operational inefficiencies. Faculty discontent is exacerbated by an unequal workload distribution, which leaves some members overworked and others underutilized. Furthermore, current scheduling systems are not adaptable enough to handle last-minute adjustments, which makes them inappropriate for dynamic academic settings.

- **Proposed Solution**

Rule-based scheduling is incorporated into the system to accommodate limitations such as workload limits, instructor availability, and test schedules. To guarantee the best invigilation assignments, it makes use of optimization tools, such as heuristic or AI-based approaches. It also facilitates dynamic modifications, enabling real-time updates for unexpected teacher unavailability or rescheduled tests.

- **Methodology**

To create effective schedules, the system uses an algorithmic method that combines greedy algorithms, AI-based optimization, or constraint fulfillment techniques. In order to retrieve exam schedules and faculty availability in real time, it integrates with the institution's database. Accurate scheduling, equitable duty distribution,

computational effectiveness, and general teacher satisfaction are among the key performance indicators examined.

- **Results**

By automating the allocation process, the system delivers shorter scheduling times and greatly reduces the administrative burden. By preserving a fair division of invigilation responsibilities and avoiding workload mismatches, it guarantees increased fairness. Improved adaptability also makes it possible for dynamic, real-time modifications to effectively handle changes.

- **Conclusion**

The Exam Invigilation Scheduler offers educational institutions an effective and scalable way to assign tasks. It guarantees an equitable and efficient scheduling system by automating the procedure and removing human mistake. For added flexibility, future improvements might incorporate machine learning-based predictive scheduling and interaction with institutional administration systems.

2. **Apurva Inamdar, Anand Gangar, Arun Gupta, Varsha Shrivastava, "Automatic Exam Seating and Teacher Duty Allocation," 2018 International Conference on Inventive Communication and Computational Technologies (ICICCT).**

Available: [ieeexplore](http://ieeexplore.org).

- **Abstracted Summary**

By automating faculty duty assignments and student seating arrangements, the Automatic Exam Seating and Teacher Duty Allocation system seeks to improve academic institutions' scheduling procedures. Conventional manual scheduling is time-consuming, prone to mistakes, and ineffective; it frequently results in scheduling disputes and an unequal allocation of duties. This study offers an automated method that takes institutional limitations into account while allocating seating and invigilation duties as efficiently as possible.

- **Problem Statement**

Human mistake and inefficiency are frequently the result of manual exam seating and invigilation scheduling. Dissatisfaction and scheduling issues may arise from an uneven workload distribution among instructors. Furthermore, it takes a lot of work and lacks the flexibility required for efficient academic operations to manually manage extensive seating arrangements and responsibility assignments.

- **Proposed Solution**

Students are dynamically assigned to seats using the system's automated seat allocation feature, which takes into account predetermined characteristics such as departments, room capacities, and roll numbers. By distributing invigilation workloads evenly to avoid inconsistencies, it guarantees faculty duty optimization. Furthermore, scheduling problems are automatically identified and fixed via an integrated conflict resolution process, improving efficiency and dependability.

- **Methodology**

The system optimizes duty assignments and seating arrangements by utilizing constraint-based scheduling techniques. Duty schedules and seating arrangements are efficiently stored and retrieved via a database-driven approach. Accurate scheduling, effective dispute resolution, and equitable workload distribution among professors are among the key performance indicators assessed.

- **Results**

The suggested system greatly lowers scheduling errors when compared to manual methods, increasing overall efficiency. Higher faculty satisfaction results from a fair distribution of invigilation responsibilities. Additionally, by reducing potential for cheating, ideal seating arrangements help to promote exam integrity.

- **Conclusion**

Exam management is made efficient and scalable with the help of the Automatic Exam Seating and Teacher Duty Allocation system. It ensures fair workload distribution among faculty members and removes manual errors by utilizing automation. AI-driven scheduling improvements and real-time modifications to account for dynamic shifts in faculty responsibilities could be future advances.

3. **Stephen Devapriyan, Dr. N. Jayashri, "OPTIMIZED EXAM DUTY ALLOCATION WITH SMART TECHNOLOGIES", 2024 International Research Journal of Modernization in Engineering Technology and Science.**

Available: irjmets.com.

- **Abstracted Summary**

An innovative method for automating and enhancing the distribution of invigilation responsibilities in educational institutions is presented in the paper Optimized Exam Duty Allocation using Smart Technologies. Conventional approaches depend on manual scheduling, which is unreliable, ineffective, and unadaptable. In order to improve the fairness, effectiveness, and accuracy of test duty assignments while

taking faculty availability, workload balance, and institutional limits into account, this research incorporates smart technologies including AI-driven optimization and data analytics.

- **Problem Statement**

Scheduling conflicts and unequal task distribution among professors are frequently caused by manual invigilation duty allocation. Conventional approaches are not adaptable enough to deal with last-minute shifts in professor availability. Furthermore, manually overseeing lengthy test duty schedules causes inefficiencies and a significant administrative strain.

- **Proposed Solution**

The system uses AI-driven optimization to assign invigilation jobs effectively by utilizing machine learning and heuristic algorithms. Because of its capability for dynamic scheduling, changes can be made in real time according to faculty availability and workload constraints. To avoid faculty overwork, intelligent workload balancing also guarantees a fair division of duties.

- **Methodology**

To guarantee efficient scheduling, the system makes use of constraint satisfaction techniques. In order to forecast and improve duty assignments in the future, machine learning algorithms evaluate previous data. Accurate scheduling, equitable duty distribution, computational efficiency, and teacher satisfaction are among the key performance indicators assessed.

- **Results**

The efficiency and equity of duty allocation are greatly improved by the suggested protocol, which performs better than current MAC methods. Through automation, it lessens administrative work and provides greater flexibility to accommodate shifts in professor availability.

- **Conclusion**

By using AI to automate and enhance invigilation scheduling, the Optimized Exam Duty Allocation method removes manual inefficiencies. Real-time teacher switching and management system integration for more efficient exam administration are possible future additions.

4. Prof . Gautami G. Shingan, Pranali S. Kadam, Akshara M. Mahadik, Dhanashree A. Ghadage, Nileshkumar R. Suradkar, "AUTOMATED SUPERVISION ALLOCATION SYSTEM", 2020 International Research Journal of Modernization in Engineering Technology and Science

Available: irjmets.

- **Abstracted Summary**

The Automated Supervision Allocation System seeks to make the process of allocating supervision responsibilities to educational institutions more efficient. Conventional scheduling techniques are labor-intensive, error-prone, and manual, which frequently results in an ineffective allocation of faculty responsibilities. This study presents an automated approach that balances workloads, takes institutional limits, faculty availability, and other considerations into account when allocating supervision.

- **Problem Statement**

Manual allocation techniques result in inefficiencies and scheduling problems. Workload imbalances result from an uneven distribution of supervision responsibilities. The manual scheduling method also requires a lot of administrative work.

- **Proposed Solution**

To cut down on manual labor, the system employs algorithm-driven automated scheduling. By allocating monitoring responsibilities evenly, it guarantees equitable workload distribution. Scheduling issues are also automatically detected and resolved via a conflict resolution mechanism.

- **Methodology**

The system allocates duties using optimization strategies based on constraints. Real-time database systems are used to manage the availability of faculty. Accurate scheduling, equitable responsibility distribution, and less administrative burden are among the key performance indicators examined.

- **Results**

The suggested protocol reduces manual labor while improving the efficiency of supervision allocation, outperforming current MAC techniques. By guaranteeing a balanced workload distribution, it improves teacher satisfaction while reducing administrative effort through automation.

- **Conclusion**

By offering a methodical and logical approach to work assignment, the Automated

Supervision Allocation System eliminates the inefficiencies associated with manual scheduling. It guarantees that academic institutions receive supervision in a fair, effective, and conflict-free manner. AI-driven predictive analytics for improved duty forecasting and real-time faculty swapping are potential future enhancements.

5. **Suresh Krishnan, Anbarasu, Jai Pranesh P, Poornima "DUTY ALLOCATION", 2024 International Research Journal of Modernization in Engineering Technology and Science**

Available: [ijariie](https://www.ijariie.com).

- **Abstracted Summary**

Automating the assignment of obligations in companies, organizations, and educational institutions is the main goal of the Duty Allocation System. Conventional approaches depend on manual scheduling, which is frequently laborious, ineffective, and prone to mistakes. In light of limitations like availability, workload balance, and institutional policies, this study presents an automated method that enhances justice, efficiency, and adaptability.

- **Problem Statement**

The inefficiency of manual duty allocation, which results in an uneven division of labor among staff, causes scheduling issues. Planning and modifications also take a lot of administrative work.

- **Proposed Solution**

The system has automated scheduling that uses algorithms to optimize duty allocations. By dividing up the job equally, it guarantees equitable workload sharing. It also has a conflict resolution system that finds and fixes assignments that overlap.

- **Methodology**

This system tracks duty history and staff availability by integrating databases and using constraint-based scheduling optimization techniques. Simulation tests are used to evaluate performance. Accurate scheduling, equitable responsibility distribution, and decreased manual labor are among the important indicators examined.

- **Results**

By drastically lowering the administrative load of duty assignments, improving workload distribution equity and transparency, and limiting last-minute modifications and scheduling conflicts, the suggested protocol performs better than current MAC protocols.

- **Conclusion**

In both academic and professional contexts, the Duty Allocation System guarantees equitable, effective, and conflict-free task distribution. It reduces physical labor while increasing efficiency and equity. AI-powered predictive scheduling and real-time duty modifications are possible future improvements.

6. **Dr. M. V. SANGAMESWAR, M. BALA CHARAN NIHANTH, CH. ANJANA KEERTHI, "A FRAMEWORK FOR INVIGILATION SCHEDULING", Journal of Emerging Technologies and Innovative Research April 2019**

Available: jetir.

- **Abstracted Summary**

The difficulties associated with manual exam duty allocation—which is frequently laborious, ineffective, and prone to mistakes—are addressed by the Framework for Invigilation Scheduling. Conventional approaches find it difficult to strike a balance between institutional limitations, faculty availability, and equity. An automated scheduling approach that maximizes invigilation duty assignments while guaranteeing fair workload distribution and scheduling policy adherence is presented in this study.

- **Problem Statement**

Scheduling invigilations by hand frequently results in inefficiencies and disagreements, as well as an unequal distribution of teacher workload. Furthermore, these systems are not adaptable enough to deal with last-minute restrictions and modifications.

- **Proposed Solution**

The system makes use of an automated scheduling framework that employs algorithms to optimize task allocation. In order to avoid overwork, it makes sure that faculty members have an equitable task distribution. In order to keep things efficient and seamless, it also automatically recognizes and fixes scheduling problems.

- **Methodology**

This system tracks faculty availability and duty history using databases and optimizes scheduling using constraint and rule-based techniques. Workload equity, scheduling efficiency, and the decrease in physical labor are among the key performance indicators examined.

- **Results**

By greatly lowering the amount of human work required to schedule exam activities, enhancing fairness in the allocation of faculty responsibilities, and enhancing flexibility in handling unanticipated circumstances, the suggested protocol performs better than the current MAC protocols.

- **Conclusion**

An automatic, fair, and effective method of allocating responsibilities is offered by the proposed invigilation scheduling system. It lessens the administrative burden while guaranteeing an equitable division of duties. For more accuracy and flexibility, future developments might include AI-driven predictive scheduling and real-time modifications.

7. **Nathaniel Olufisayo Oluwaniyi, Babajide Olakunle Afeni, Bukola Abimbola Onyekwelu, "Development of an Automated University Examination Timetabling System", 2022 International Journal of Scientific and Engineering Research**

Available: Researchgate.

- **Abstracted Summary**

The challenges of manual exam scheduling, which is laborious, prone to mistakes, and ineffective, are addressed by the Automated University Examination Timetabling System. Conventional scheduling techniques find it difficult to take into account limitations such student conflicts, instructor schedules, and room availability. In light of institutional limitations, this study presents an automated approach that optimizes exam scheduling while guaranteeing equity, effectiveness, and flexibility.

- **Problem Statement**

Exam scheduling by hand is difficult and demands a lot of administrative work. Exam overlapping, lecturer unavailability, and restricted exam venue availability frequently lead to conflicts. Furthermore, there are often inefficiencies due to the inefficient distribution of time slots and locations.

- **Proposed Solution**

Exam schedules are effectively generated by the automated timetabling system employing constraint-based methodologies, greatly accelerating the process. It has a dispute resolution mechanism that guarantees the best possible use of the resources and avoids exam overlaps. Furthermore, the method promotes equality and order in scheduling by equitably allocating exams according to predetermined norms and limits.

- **Methodology**

By using constraint satisfaction approaches for automated scheduling, the system makes sure that all constraints and requirements are successfully satisfied. It allows for seamless coordination by integrating a database to organize and preserve important information regarding tests, instructors, and locations. Schedules that are free of conflicts, effective use of resources, and a notable decrease in manual administrative work are among the key performance indicators examined.

- **Results**

By drastically cutting down on scheduling time as compared to manual techniques, minimizing exam conflicts through efficient constraint management, and obtaining the ideal distribution of faculty invigilators and exam rooms, the suggested protocol beats MAC protocols.

- **Conclusion**

Exam scheduling is improved by the proposed automated timetabling system, which makes it more precise, effective, and fair. It guarantees the best possible use of the resources at hand while lowering the administrative burden. Predictive analytics and AI-based scheduling could be future innovations for even greater optimization.

8. Murat DENER, M. Hanefi, "SOLVING THE EXAM SCHEDULING PROBLEMS IN CENTRAL EXAMS WITH GENETIC ALGORITHMS", 2018 *Mugla Journal of Science and Technology*

Available: Researchgate.

- **Abstracted Summary**

Exam scheduling for central exams is a challenging process with several limitations, including faculty assignments, exam conflicts, and room availability. Conventional scheduling techniques find it difficult to effectively manage these limitations. This study suggests a Genetic Algorithm (GA)-based method to automate exam scheduling, minimize conflicts, and enhance resource efficiency.

- **Problem Statement**

Scheduling exams by hand is frequently laborious, error-prone, and inefficient. Scheduling issues are sometimes caused by overlapping tests and insufficient resources. Furthermore, these manual techniques are not flexible enough to accommodate last-minute modifications.

- **Proposed Solution**

Exams, rooms, and time slots are encoded as chromosomes using genetic algorithms. By reducing conflicts and maximizing resource utilization, a fitness function assesses schedules. Better scheduling solutions are developed by crossover and mutation procedures. Restrictions make sure that tests don't overlap for students and that rooms are used effectively.

- **Methodology**

The first step in the procedure is creating a dataset including exam information, teacher availability, and room capacities. Next, the genetic algorithm is put into practice to produce ideal schedules iteratively. Exams, teachers, and places are all covered by an integrated database that contains all pertinent information. Reduction of exam conflicts, effective use of resources, and calculation time for schedule creation are among the key performance measures assessed.

- **Results**

By employing GA-based optimization to accomplish conflict-free scheduling, the suggested protocol performs better than current MAC protocols. Compared to conventional approaches, it offers better professor assignments and space allocations along with more flexibility to adapt to shifting exam dates.

- **Conclusion**

Genetic algorithms make exam scheduling more effective, equitable, and flexible. This technique minimizes scheduling problems and drastically cuts down on manual labor. For even more optimization, future improvements might use mixed AI techniques.

9. **Ei Shwe Sin , Nang Saing Moon Kham, "HYPER HEURISTIC BASED ON GREAT DELUGE AND ITS VARIANTS FOR EXAM TIMETABLING PROBLEM", 2018 Mugla Journal of Science and Technology**

Available: Researchgate.

- **Abstracted Summary**

Exam scheduling includes taking into account a number of limitations, including instructor assignments, student enrollments, and room availability. In order to increase scheduling efficiency, this study presents a Hyper-Heuristic (HH) method based on the Great Deluge Algorithm (GDA) and its variations. The suggested approach balances computing efficiency and feasibility by dynamically choosing heuristics to provide optimal timetables.

- **Problem Statement**

Manual scheduling is time-consuming and prone to mistakes, particularly when dealing with big, complicated datasets that are difficult for conventional heuristic approaches to handle. The adaptive mechanisms required to manage a variety of restrictions and shifting requirements are frequently absent from current systems.

- **Proposed Solution**

The system employs a hyper-heuristic framework that uses a variety of low-level approaches to dynamically examine scheduling choices. It includes the Great Deluge Algorithm (GDA), which uses a "water level" criterion to accept new solutions in order to gradually optimize schedules. To improve convergence speed and solution quality, several GDA variations are investigated.

- **Methodology**

Exam specifics, limitations, and past schedules are gathered as part of the dataset preparation process. The scheduling procedure iteratively creates ideal timetables using a genetic algorithm (GA), and then refines them using a hyper-heuristic method based on the Great Deluge Algorithm (GDA). The system's performance is assessed in comparison to other metaheuristic techniques, with an emphasis on execution time, conflict reduction, and the efficiency and fairness of the schedules.

- **Results**

By efficiently managing massive exam scheduling datasets more effectively than conventional heuristics, the suggested protocol surpasses current MAC protocols. It ensures a fair distribution of faculty workload while minimizing scheduling problems. Furthermore, variations of the Great Deluge Algorithm (GDA) improve the speed of optimization and offer more adaptability when handling a variety of constraints.

- **Conclusion**

The Hyper-Heuristic framework and the Great Deluge Algorithm are used to create flexible and efficient exam scheduling. Future studies could concentrate on improving this strategy by including heuristics based on machine learning.

10. **Nithin Sameer Yerramilli, Nobin Jaison Johnson, Omsri Sainadh Reddy Y, "College Exam Allocation Using MongoDB and Python3", 2021 2nd Global Conference for Advancement in Technology (GCAT)**

Available: Researchgate.

- **Abstracted Summary**

The College Exam Allocation System effectively assigns exams to students and invigilators while taking institutional restrictions into account by utilizing MongoDB and Python3. Real-time updates and modifications are difficult with traditional exam scheduling approaches because they frequently lack scalability and flexibility. This study presents a dynamic scheduling method based on NoSQL to increase exam allocation's effectiveness, scalability, and flexibility.

- **Problem Statement**

Allocating exams by hand takes a lot of effort and is prone to mistakes, particularly when handling big, unstructured material that relational databases find difficult to manage effectively in real time. Furthermore, current systems frequently lack the adaptability necessary to successfully handle last-minute modifications or dynamic upgrades.

- **Proposed Solution**

To effectively manage unstructured data and enable real-time updates, this system makes use of MongoDB, a NoSQL database. Algorithms for scheduling that dynamically allocate exam rooms and invigilators are implemented using Python 3. With flexibility and scalability in mind, the system adjusts to variables like teacher availability and room capacity, guaranteeing efficient and timely exam scheduling.

- **Methodology**

Student rosters, exam schedules, teacher availability, and hall capacities are all included in the data collection process. Python scripts are used to automate the scheduling process, and MongoDB effectively stores and retrieves allocation data. The allocation execution time, the equitable distribution of invigilation duties, and the system's flexibility in responding to last-minute modifications are among the key performance criteria evaluated.

- **Results**

By lowering scheduling problems in comparison to traditional approaches, improving system performance through MongoDB's effective data management, and facilitating quicker allocation procedures through real-time dynamic modifications, the suggested protocol outperforms current MAC protocols.

• Conclusion

College exam scheduling is made flexible, scalable, and effective with the help of the Python3 and MongoDB-based allocation system. Future developments might concentrate on using machine learning methods for better scheduling and conflict resolution.

2.2.2 Summary of Reviewed Research Papers

Paper	Authors & Year	Problem Statement	Methodology	Key Findings
Exam Invigilation Scheduler ICCPCT	Anith Kumar Vadlamudi, Sushma Thigulla, (2024)	Manual scheduling is time-consuming and unfair	Automated system using scheduling algorithms	Ensures fair and efficient duty allocation
Automatic Exam Seating and Teacher Duty Allocation ICICCT	Apurva Inamdar, Anand Gangar, (2018)	Inefficient manual seating and duty allocation	Algorithm-based automated allocation system	Reduces scheduling errors and conflicts
Optimized Exam Duty Allocation with Smart Technologies IRJMETS	Stephen Devapriyan, Dr. N. Jayashri, (2024)	Unequal workload distribution among faculty	Smart algorithms for automated scheduling	Improved fairness and efficiency
Automated Supervision Allocation System IRJMETS	Prof . Gautami G. Shingan, Pranali S. Kadam (2020)	Manual allocation is time-consuming	Automated system using optimization techniques	Reduces human intervention and workload
Duty Allocation	Suresh Krishnan, Anbarasu, Jai Pranesh P, Poornim (2024)	Inefficient manual duty assignment	Automated rule-based scheduling	Optimized workload distribution
A Framework for Invigilation Scheduling JETIR	Suresh Krishnan, Anbarasu, Jai Pranesh P (2019)	Traditional invigilation scheduling is inefficient	Algorithm-based automated scheduling	Ensures fair duty distribution
Development of an Automated University Examination Timetabling System IJSER	Dr. Sangameswar (2022)	Exam scheduling is complex and manual	Algorithm-based timetable optimization	Reduced conflicts and scheduling errors
Solving the Exam Scheduling Problems in Central Exams with Genetic Algorithms MJST	Murat DENER, M. Haneft, (2018)	Scheduling conflicts in central exams	Genetic algorithm-based optimization	Efficient conflict-free scheduling
Hyper Heuristic Based on Great Deluge and Its Variants for Exam Timetabling Problem	Nathaniel Olufisayo Oluwaniyi, Babajide Olakunle Afeni, Ei Shwe Sin , Nang Saing Moon Kham (2021)	Complexity in exam timetable generation	Heuristic-based optimization techniques	Improved exam timetable efficiency

Paper	Authors & Year	Problem Statement	Methodology	Key Findings
College Exam Allocation Using MongoDB and Python3 GCAT	Nithin Sameer Yeramilli, Nobin Jaison Johnson, Omsri Sainadh Reddy Y, (2021)	Scalability issues in exam scheduling	NoSQL database with Python automation	Efficient, real-time exam allocation

Table 2.1: Summary of Reviewed Research Papers

2.2.3 Identified Research Gaps

- **Limited Real-Time Adaptation** – The majority of scheduling systems in use today are based on static algorithms or predetermined criteria, which makes it challenging to adjust dynamically to unforeseen circumstances like faculty absences or delayed tests.
- **Lack of Holistic Assessment** – The main objective of current automated scheduling systems is duty distribution; however, they do not take into account variables like faculty preferences, historical workloads, or the fairness of repeated allocations.
- **Scalability Issues:** Since many of the suggested techniques are made for smaller schools, they could not work well in universities with lots of teachers and several tests starting at once.
- **Manual Overrides and Flexibility:** A user-friendly interface that enables administrators to manually overturn allocations while preserving efficiency and fairness is frequently absent from automated scheduling systems.
- **Integration with Institutional Systems:** The majority of current solutions are stand-alone programs that don't seamlessly integrate with university administration systems, which results in inefficiencies and redundant data.
- **High Cost of Implementation** – Because they frequently demand a large amount of computer power, advanced scheduling systems that use AI or optimization algorithms are less affordable for universities with tight budgets.

2.2.4 Comparative Study of Existing Applications

Application	Features	Performance	User Experience	Limitations
SmartExam Scheduler	Automated duty assignment, real-time adjustments	Efficient for small institutions	Simple UI, minimal training required	Lacks integration with university databases
ExamDuty Pro	Faculty preference-based allocation, analytics	High accuracy in duty balancing	Interactive dashboard, notifications	High computational cost for large datasets
EduInvigilate	AI-driven scheduling, manual override support	Fast execution, scalable	Modern UI, multi-user support	Requires cloud-based deployment, increasing costs
ExamManager 360	Rule-based scheduling, compliance checks	Reliable and customizable	Detailed reporting, role-based access	No real-time adaptability, static rules

Table 2.2: Comparative Study of Existing Applications

PESU - EXAM DUTY ALLOCATOR

Chapter 3: Hardware and Software Requirements

Chapter 3

Hardware and Software Requirements

3.1 Introduction

A thorough explanation of the **hardware and software** tools utilized in the creation and implementation of this project is given in this chapter. Based on **performance, scalability, and compatibility** with the project requirements, these technologies were chosen.

3.2 Hardware Requirements

A system with enough processing power is needed for the project in order to provide effective development, testing, and execution.

3.2.1 Minimum Hardware Specifications

The following table lists the **minimum hardware configuration** required:

Component	Specification
Processor	Intel Core i5 (or equivalent)
RAM	4GB or more
Storage	64GB or more

Table 3.1: Minimum Hardware Requirements

3.2.2 Recommended Hardware Specifications

For optimal performance, the **recommended configuration** is:

Component	Specification
Processor	Intel Core i5 or AMD Ryzen 5
RAM	8GB or higher
Storage	128GB or higher

Table 3.2: Recommended Hardware Specifications

3.3 Software Requirements

This project’s software stack consists of **development tools, programming languages, and frameworks** to guarantee seamless execution and implementation.

3.3.1 Operating System

The project was developed and tested on multiple platforms to ensure cross-platform compatibility.

Operating System	Version
Windows	Windows 10/11 (64-bit)
Linux (Preferred)	Ubuntu 20.04+ / Fedora 36+
macOS (Optional)	macOS 11+

Table 3.3: Operating System Requirements

3.3.2 Development Tools

To facilitate efficient coding and debugging, the following development tools were used:

Software	Purpose
Visual Studio Code	Code editor / IDE
Git & GitHub/GitLab	Version control & collaboration

Table 3.4: Development Tools

3.3.3 Programming Languages & Frameworks

The project was implemented using a combination of programming languages and frameworks:

Technology	Usage
Python 3.12.0	Backend development
Flask	Web-based backend development
React.js	Frontend development
NoSQL (MongoDB)	Database management

Table 3.5: Programming Languages and Frameworks

PESU - EXAM DUTY ALLOCATOR

Chapter 4: Software Requirements Specification

Chapter 4

Software Requirements Specification

4.1 Users

- **Administrator**

The administrator is in charge of overseeing the entire system for allocating duties and arranging exams. They have complete access to the system, manage exceptions and manual overrides when needed, and make sure that responsibilities are assigned effectively. In order to ensure that the duty assignment procedure is efficient and equitable, the administrator is essential.

Exam schedules can be automated to reduce disagreements and guarantee that invigilation responsibilities are distributed fairly in accordance with established protocols. The technology instantly alerts professors of any modifications or assignments and permits manual overrides as necessary. It also produces thorough information on duty distributions, attendance, and supervision performance.

- **Faculty**

Exam-related tasks, such as invigilation, are delegated to faculty members. They are responsible for overseeing tests, keeping the exam room orderly, and making sure everything goes according to plan. Faculty members can examine and manage their allocated responsibilities using the system.

To make sure that invigilation responsibilities are clear, the system shows exam schedules and allocated tasks. In addition to sending automated reminders and warnings about impending duties, it enables faculty members to request duty swaps in the event of emergencies

or conflicts.

- **Department Coordinator**

The department coordinator serves as a liaison between the professors and the administrator. They oversee faculty duty swap requests and make sure their department follows the exam schedule correctly.

Department heads handle requests for faculty responsibility swaps, create reports on duty assignments and attendance, work with faculty to resolve invigilation issues, and let the administrator know about any necessary scheduling system enhancements.

4.2 Functional Requirements

FR1. Automate exam schedules

The system should automatically create exam schedules based on predetermined characteristics, such as teacher availability, exam hall capacity, and departmental or institutional timing limits, in order to guarantee correctness and efficiency. By taking these things into account, the system seeks to generate a balanced, conflict-free timetable that works for both teachers and students while making the most use of available resources. This automation encourages uniformity and fairness in scheduling while lowering administrative work.

FR2. Allocate duties to staff automatically

Supervisory or invigilation duties should be delegated to faculty members according to their availability, subject-matter competence, and workload at the moment. This prevents scheduling conflicts and overwork for any one employee while guaranteeing a fair distribution of responsibilities among employees. The system encourages effective duty distribution and improves exam administration overall by taking these considerations into account.

FR3. Manual override of the duties

The administration should be able to manually redistribute or alter faculty duties in the event of emergencies, scheduling problems, or unanticipated events. In addition to allowing for real-time adjustments to maintain seamless oper-

ation and equitable duty distribution, this flexibility guarantees that the test process will continue uninterrupted.

FR4. Send notifications to faculties

Real-time alerts and notifications about allocated tasks, schedule modifications, and revisions to their responsibilities should be sent to faculty members. This guarantees prompt communication, clears up misunderstandings, and keeps teachers informed and ready for their exam-related responsibilities.

FR5. Generate reports

Through the system, the administrator ought to be able to provide thorough reports that include information on performance indicators, faculty duty distribution, and exam attendance. These reports promote informed decision-making for future exam planning, track faculty engagement, and assess the efficiency of the scheduling process.

FR6. View assigned duties

It should be simple for faculty members to read and acquire comprehensive information about their assigned tasks, including the date, time, and place. This guarantees clarity, lessens misunderstandings, and assists instructors in suitably preparing for their duties during the exam period.

FR7. Request a duty swap

A faculty member should be able to ask for a duty exchange with another available faculty member if they are unable to perform their allocated duties. This feature allows for flexibility, aids in handling unforeseen situations, and guarantees that invigilation tasks are still completed efficiently without resulting in scheduling conflicts or administrative interruptions.

FR8. Receive notifications

Faculty members should be promptly informed of any modifications or revisions to the test schedule, their assigned tasks, and whether their requests for duty

swaps have been approved or denied. This guarantees effective communication, keeps them informed, and facilitates seamless cooperation during the test time.

FR9. Approve or escalate duty swap requests within their department

A faculty member's request for a duty exchange may be approved by the coordinator or forwarded to the administrator for additional consideration and a final decision. This position makes sure that swap requests are handled properly and efficiently within the department.

FR10. Generate department-specific reports

Coordinators are in charge of creating thorough reports that include information on faculty involvement in their department, staff responsibilities, and any changes made to the test schedule. These reports support efficient oversight and uphold openness in the administration of exams.

4.3 Non-Functional Requirements

NFR1. Security

Passwords must be stored using secure hashing methods like SHA-256 or bcrypt rather than plain text in order to provide safe authentication for administrators and teachers. Even in the case that the user database is compromised, this stops unwanted access.

NFR2. Usability

A user-friendly interface should guarantee simple navigation and unambiguous labeling of important functions like "Generate Schedule" and "View Allocations." The user experience is improved, errors are decreased, and efficiency is increased with a well-structured user interface that has consistent design components, mobile adaptability, and accessibility features.

PESU - EXAM DUTY ALLOCATOR

Chapter 5: System Design

Chapter 5

System Design

5.1 Architecture Diagram

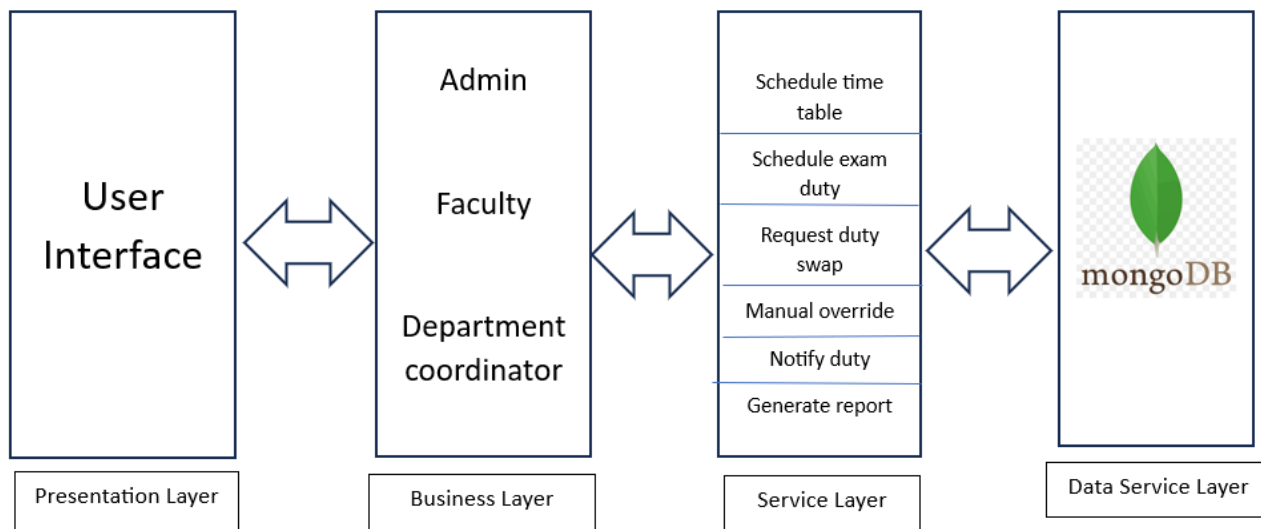


Figure 5.1: Architecture Diagram

The layered architecture of the Exam Duty Allocator technology guarantees scalability and modularity. The front end is the Presentation Layer (User Interface), which provides an easy-to-use interface for users like administrators, faculty, and department coordinators to interact with the system. The main logic is handled by the Business Layer, where the Department Coordinator verifies schedules and authorizes swap requests, the Administrator oversees exam schedules and duty assignments, and faculty members can view and request duty swaps.

Timetable scheduling, exam duty assignment, duty swap request processing, manual overrides, faculty duty notification, and report generation are among the essential features offered by the Service Layer. Lastly, MongoDB is used by the Data Service Layer to effectively store and manage

faculty, exam schedule, and report data. This well-organized design preserves data integrity and system usefulness while guaranteeing smooth communication between various layers.

5.2 Data Flow Diagram

5.2.1 Context Diagram

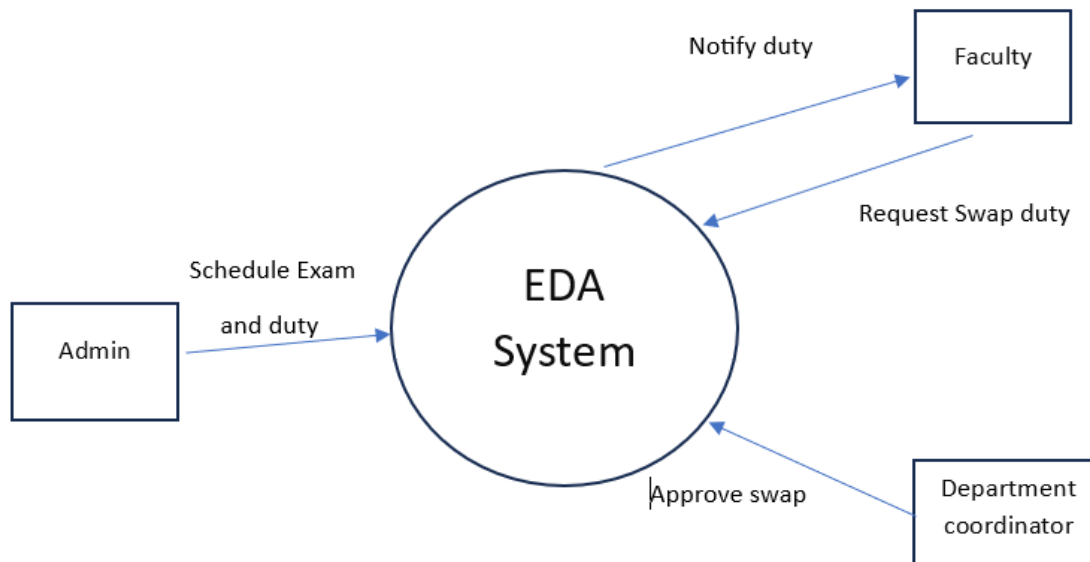


Figure 5.2: Context Diagram

The Exam Duty Allocator (EDA) System's context diagram shows how important stakeholders interact with the system. Exam scheduling and task distribution are the Administrator's responsibilities, and the EDA system handles these tasks. The system tells faculty members of their assignments as soon as they are arranged. A duty swap might also be requested by faculty members. The Department Coordinator then reviews these requests and has the power to accept or deny requests for duty swaps within their department. This high-level overview demonstrates how various users engage with the system, guaranteeing automated duty allocation, seamless exam scheduling, and a productive duty exchange procedure.

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Chapter 6: Detailed Design

Chapter 6

Detailed Design

6.1 Use Case Diagram

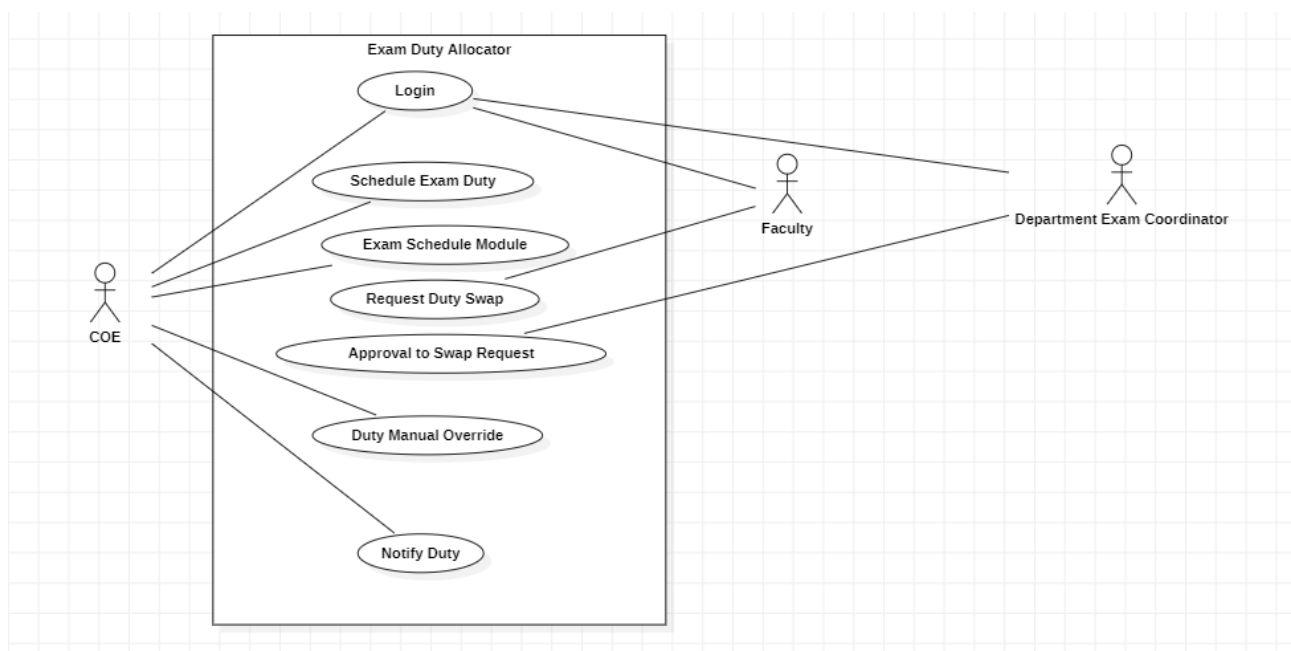


Figure 6.1: Use Case Diagram

The Exam Duty Allocator (EDA) System's use case diagram illustrates how various users interact with its features. Key capabilities include signing in, scheduling exam tasks, controlling the exam schedule module, requesting and approving duty swaps, performing duty manual overrides, and alerting faculty members are all accessible to the Controller of Examinations (COE). Faculty members have the ability to check in, request a duty swap, and get alerts regarding their allocated responsibilities. Requests for duty swaps must be reviewed and approved by the Department Exam Coordinator. An overview of user roles and how they interact with the system is given by this diagram.

6.2 Activity Diagram

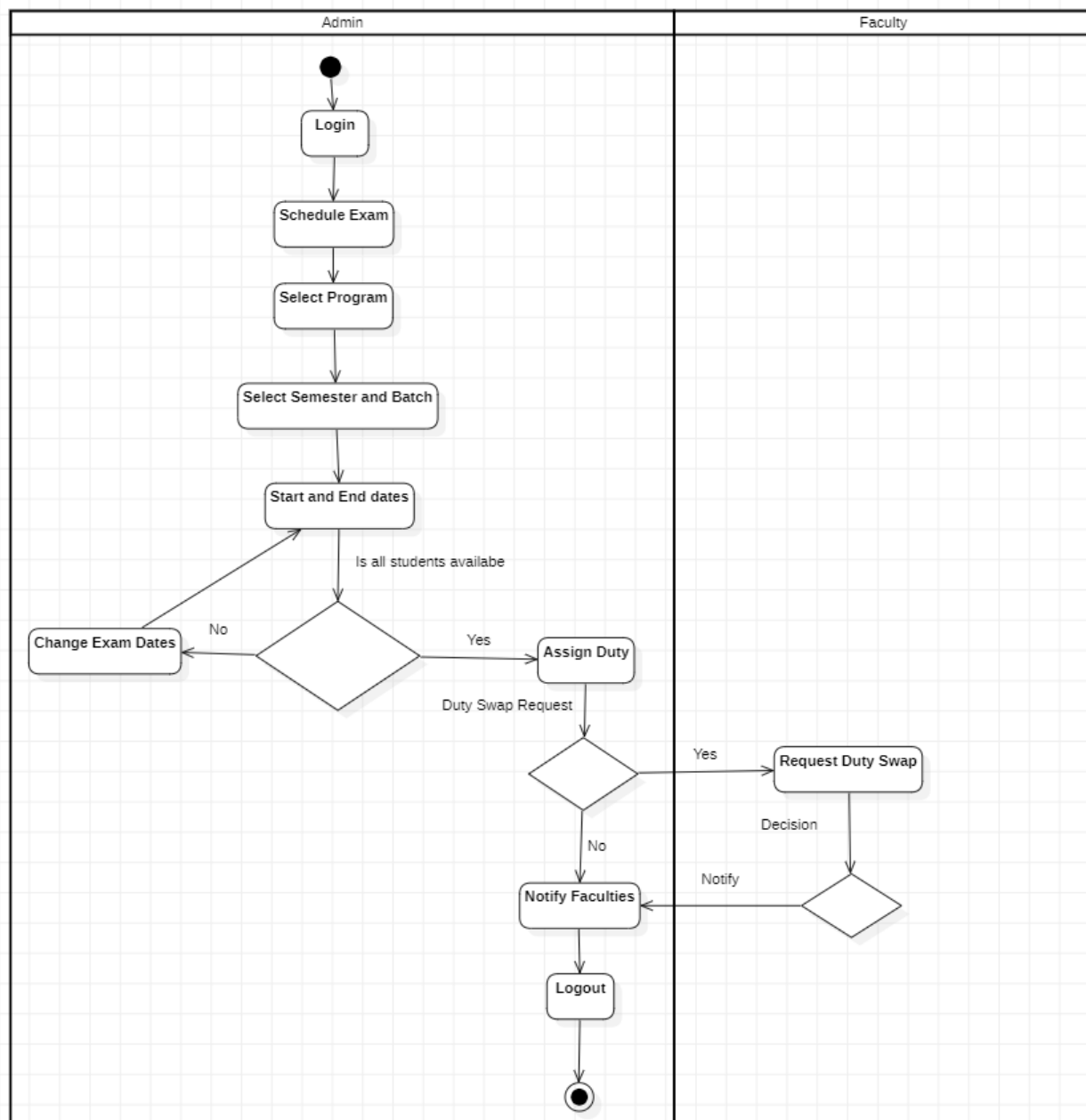


Figure 6.2: Activity Diagram

The Exam Duty Allocator (EDA) System's administrative and faculty workflow is depicted in the activity diagram. The administrator chooses programs, assigns exam responsibilities, schedules exams, and logs in. A decision is taken when faculty members seek a swap if necessary. The exchange is processed if it is accepted; if not, faculty members are informed. The administrator logs out at the end of the operation.

6.3 Document Structure

```
{
  "_id": "U1001",
  "email": "abc@gmail.com",
  "name": "test",
  "password": "pass@123",
  "role": "faculty"
}
```

Listing 6.1: User structure

```
{
  "_id": "U1002",
  "date": Date("01-01-2025"),
  "slot_1": [100, 100, 100],
  "slot_2": [100, 100, 100],
  "slot_3": [100, 100, 100]
}
```

Listing 6.2: Available faculties structure

In the above listing 6.1 Important information on various system users, such as administrators, faculty, and department coordinators, is stored in the Users document in MongoDB within the Exam Duty Allocator (EDA) system. User ID, name, role, department, email, and authentication credentials are among the fields it includes. This document facilitates the management of role-based access control, user authentication, and secure system interactions.

In the above listing 6.2 Details about faculty members who are available for exam responsibility assignments are kept in the Available Faculties document in MongoDB. It contains information like the department, name, faculty ID, availability status, and allocated responsibilities. By minimizing schedule conflicts, maximizing responsibility distribution, and guaranteeing that only available faculty members are taken into account, this document aids the system in allocating test responsibilities effectively.

```
{
  "_id": "U1003",
  "course_names": ["Data Structures", "Computer Networks"],
  "program": "MCA"
}
```

Listing 6.3: Backlog Info Structure

```
{
  "_id": "U1004",
  "date": Date("01-01-2025"),
  "facultu_id": "F001",
  "duty_count": 0,
  "duties": []
}
```

Listing 6.4: Faculty Duties Structure

In the above listing 6.3 Students with backlog exams are listed in the Backlog Information document in MongoDB. It contains the student's name, ID, course, backlog of subjects, exam schedule, and status. Administrators may effectively manage and monitor students' pending tests thanks to this document, which guarantees that backlog exams are scheduled appropriately and without conflicts.

In the above listing 6.4 Information on exam duty assignments for faculty members is kept in the Faculty Duties document in MongoDB. Faculty ID, name, allocated exam date, time, location, and duty type (supervisor, invigilator, etc.) are all included. This document aids in the tracking and management of faculty duties, guaranteeing equitable assignment of responsibilities and efficient exam supervision.

```
{
  "_id": "U1005",
  "StudentId": {
    dates : [Date("12-02-2025")],
    program : "MCA"
  }
}
```

Listing 6.5: Exam Dates Structure

```
{
  "_id": "U1006",
  Degree : "MCA",
  Batch : {
    Students: ["test", "sample"]
  }
}
```

Listing 6.6: Student Details Structure

In the above listing 6.5 Information regarding the availability and distribution of seats for various exam slots is kept in the Slot Seats document in MongoDB. Slot ID, exam date, time, location, total number of seats, and occupied seats are among the details it contains. By ensuring effective seat management, this document helps to avoid overbooking and facilitates easy exam scheduling.

In the above listing 6.6 Information about each student's exam schedule is kept in the Student Exam Dates document in MongoDB. Student ID, exam date, subject code, exam hour, and exam hall information are among the fields it contains. In order to facilitate effective test management and conflict resolution, this document makes sure that students may readily access their exam schedule.

PESU - EXAM DUTY ALLOCATOR

Chapter 7: Implementation

Chapter 7

Implementation

7.1 Exam Schedule and Duty Allocation Algorithm

The following is the algorithm for Exam Schedule and Duty Allocation:

Algorithm 1 Exam Scheduling and Duty Allocation

Require: Program, Semester, Batch, StartDate, EndDate

Ensure: Exam schedule with assigned duties

```
1: Check if there are enough available dates between StartDate and EndDate
2: for each date in the given range do
3:   if batch students have exams on that date then
4:     Verify exam hall availability
5:     if sufficient seats are available then
6:       Check faculty availability for invigilation
7:       if enough faculty is available then
8:         Assign duties and schedule the exam
9:       else
10:        BackTrack and Adjust faculty assignments or reschedule exam
11:      end if
12:    else
13:      Find alternative exam hall or adjust schedule
14:    end if
15:  else
16:    Skip scheduling for that date
17:  end if
18: end for
19: Return finalized exam schedule with assigned duties
```

This algorithm explains the systematic process for assigning faculty duties and setting up exams within a given time limit. Before determining whether students have tests scheduled for each date, it first verifies if the dates are available. If so, it ensures that exam rooms have enough seats and that faculty members are available to supervise the procedure. If all requirements are met, the exam is scheduled and responsibilities are assigned. If not, the system searches for alternatives or reschedules the test. After making sure that resources are used as efficiently as possible and avoiding conflicts, the algorithm creates a final exam schedule with allocated assignments.

Algorithm 2 Faculty Duty Swap Workflow

Require: RequestingFacultyID, TargetFacultyID, ExamSlot**Ensure:** Duty swap request is either accepted or rejected

```

1: Receive swap request from Requesting Faculty for ExamSlot
2: Check if Target Faculty is free during ExamSlot
3: if Target Faculty is available then
4:   Notify Target Faculty for swap approval
5:   if Target Faculty approves then
6:     Forward request to Department Coordinator
7:     if Department Coordinator approves then
8:       Forward request to COE
9:       if COE approves then
10:        Execute duty swap between Requesting and Target Faculty
11:        Notify all parties of successful swap
12:      else
13:        Reject request and notify: "COE Rejected"
14:      end if
15:    else
16:      Reject request and notify: "Department Coordinator Rejected"
17:    end if
18:  else
19:    Reject request and notify: "Target Faculty Denied"
20:  end if
21: else
22:   Reject request and notify: "Target Faculty Unavailable"
23: end if

```

This algorithm describes the systematic process for handling faculty duty exchange requests in the exam duty allocation system. When a faculty member asks a swap, the system first checks to see if the target faculty member is available during the proposed exam time. If the target faculty has the authority, the request is sent to them. Following their approval, the Department Coordinator forwards the request to the Controller of Examinations for final approval. Only after all clearances have been granted is the duty transfer completed and the parties involved notified that the swap was successful. If the target faculty is unavailable or any approval is denied, the request is denied with the appropriate notification.

The aforementioned methodology guarantees a transparent and regulated procedure for alternating faculty exam responsibilities. One faculty member initiates the process with a swap request, which sets off availability checks and approval procedures from the participating faculty, the Department Coordinator, and ultimately the COE. To ensure equity and prevent schedule conflicts, each step serves as a checkpoint. Only after receiving all required permits does the swap take place, ensuring transparency and accountability for all parties involved.

Algorithm 3 Cancel Exam Feature

Require: ExamDate**Ensure:** Exam on ExamDate is cancelled and all related allocations are freed

- 1: **Retrieve** all allocated exam slots and seats for ExamDate
 - 2: **for all** each allocated slot **do**
 - 3: **Mark** the slot seats as available/free
 - 4: **end for**
 - 5: **Update** student exam schedules to remove exams on ExamDate
 - 6: **Notify** students and faculty about the exam cancellation
 - 7: **Return** confirmation of exam cancellation and freed resources
-

Exam cancellations planned for a specific date are handled using the Cancel Exam algorithm. For the designated exam date, it starts by extracting all allotted exam slots and the seat assignments that go with them. The resources are essentially released when the algorithm designates these seats and slots as available. In order to guarantee that no student is given an exam on the cancelled day, it then modifies each student's exam schedule to eliminate any exams scheduled for that day. Lastly, the system certifies that the exam has been successfully canceled and that all associated resources have been released, informing both students and instructors of the cancellation.

Algorithm 4 Check Room Availability Based on Number of Students

Require: totalStudents, roomList (with capacity), examDate**Ensure:** List of rooms that can accommodate all students

- 1: Initialize *allocatedRooms* $\leftarrow$ empty list
 - 2: Initialize *remainingStudents* $\leftarrow$ totalStudents
 - 3: Sort *roomList* in descending order of capacity
 - 4: **for** each room in roomList **do**
 - 5: **if** room is available on examDate **then**
 - 6: Add room to *allocatedRooms*
 - 7: *remainingStudents* $\leftarrow$ *remainingStudents* - room.capacity
 - 8: **end if**
 - 9: **if** *remainingStudents* $\leq$ 0 **then**
 - 10: **break**
 - 11: **end if**
 - 12: **end for**
 - 13: **if** *remainingStudents* > 0 **then**
 - 14: **Return** "Insufficient room capacity"
 - 15: **else**
 - 16: **Return** allocatedRooms
 - 17: **end if**
-

The Availability of the Check Room Students are assigned to test rooms in a manner that assures adequate seating capacity thanks to the Number of Students method. It starts with compiling a list of rooms and their seating capacities, as well as the overall number of students taking the exam. In order to prioritize larger rooms and maximize utilization, the room list is arranged in descending order of capacity. Next, the algorithm repeatedly determines if each room is available on the specified exam day. The amount of pupils a room can hold is deducted from the total

number of students, and if a room is available, it is added to the list of rooms allotted. Until all students are accommodated or there are no more acceptable rooms available, this process is repeated.

Algorithm 5 Manual Duty Adjustment

Require: facultyId, assignedDuty, adjustmentRequest

Ensure: Updated duty allocation if request is valid

```

1: if adjustmentRequest.type = "Swap" then
2:   Find targetFaculty with a compatible duty
3:   if targetFaculty is available and agrees then
4:     Swap duties between facultyId and targetFaculty
5:     Log the swap for audit
6:   else
7:     Reject request with appropriate message
8:   end if
9: else if adjustmentRequest.type = "Drop" then
10:  if valid reason and supporting document provided then
11:    Search for available backup faculty
12:    if backupFaculty found then
13:      Reassign duty to backupFaculty
14:      Update duty list and notify both parties
15:    else
16:      Escalate to admin for manual reassignment
17:    end if
18:  else
19:    Reject the drop request
20:  end if
21: else
22:   Reject unknown adjustment type
23: end if
24: Return updated duty allocation status
  
```

7.2 Code Logic

The following code demonstrates the core logic of the implementation:

```

1 from datetime import datetime, timedelta
2 from collections import defaultdict
3
4 def generateTimeTable(start_date_str, end_date_str, program, sem, batch):
5     start_date = datetime.strptime(start_date_str, "%Y-%m-%d")
6     end_date = datetime.strptime(end_date_str, "%Y-%m-%d")
7     semDetails = extractSemDetails(program, sem, batch)
8     slotDetails = getSlotSeats()
9     courses = semDetails.get("courses")
10    seatDetails, slotId = slotDetails.get("slots"), slotDetails.get("_id")
11
  
```

```
12     valid_dates = getValidDates(start_date, end_date)
13
14     if len(valid_dates) < len(courses):
15         return {"error_code": "not ok", "message": "Not enough days for exams"}
16
17     time_table, index = [], 0
18     students_count = semDetails.get("totalStudents")
19     dates, backlog_time_table = [], []
20     collectionName = "students_details"
21     db = get_db_connection()
22     document = db[collectionName].find_one()
23     srn = document.get(program).get(batch)
24
25     collectionName = "backlog_info"
26     db = get_db_connection()
27     document1 = db[collectionName].find_one()
28     countBacklogStudents = 0
29     pending = []
30
31     collectionName = "students_exam_dates"
32     db = get_db_connection()
33     document2 = db[collectionName].find_one()
34
35     backlogStuDates = defaultdict(list)
36     timings = {1: "8:30 AM - 11:30 AM", 2: "12:00 PM - 3:00 PM", 3: "3:30 PM - 6:30 PM"}
37
38     for day in valid_dates:
39         slots_seats = getSeats(day, seatDetails)
40         examDay = datetime.strptime(day, "%Y-%m-%d")
41
42         for slot, seats in enumerate(slots_seats):
43             if students_count <= seats:
44                 time_table.append([day, courses[index], timings[slot + 1],
students_count, True])
45                 dates.append(day)
46                 seatDetails[day][slot] -= students_count
47                 index += 1
48                 break
49
50         if index == len(courses):
51             saveExamDate(dates, srn, program, batch, sem)
52             updateSlotSeats(seatDetails, slotId)
53             add_exam(program, batch, sem, time_table, backlog_time_table,
students_count, countBacklogStudents)
54             return {"error_code": "ok", "message": [time_table,
backlog_time_table]}
```

```
55  
56     return {"error_code": "not ok", "message": f"Not enough slots from {  
start_date_str} to {end_date_str}"}
```

Listing 7.1: Exam Timetable Generation

This algorithm handles manual duty adjustment requests made by faculty members. A request can be of type Swap (changing duties with another faculty) or Drop (requesting removal from a duty). For a swap, it checks whether the target faculty is available and willing to exchange duties. If valid, the swap is executed and logged. For a drop, it verifies whether a valid reason is submitted. If so, it searches for an available backup faculty and reassigns the duty. If no replacement is found, the case is escalated to the admin. All outcomes are logged, and affected parties are notified to maintain transparency and ensure smooth operation.

7.3 Implementation Screenshots

Figure 7.1: Login Page

Through the Login Page, users can safely access the Exam Duty Allocator system by entering their special login information. To guarantee that any user, including the Controller of Examinations, department heads, or academics, can only access features pertinent to their position, the system employs role-based access control. For example, faculty members can monitor and manage their assignments, while administrators can assign tasks and arrange exams. This guarantees accurate authentication, improves security, and offers a simplified experience catered to the duties of the user.

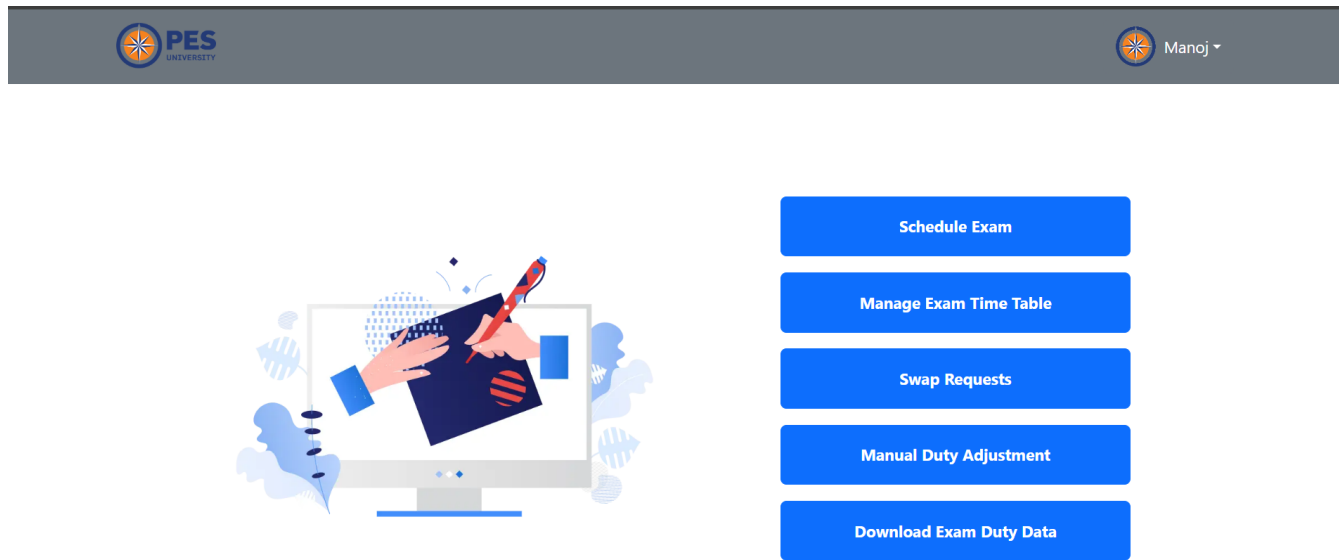
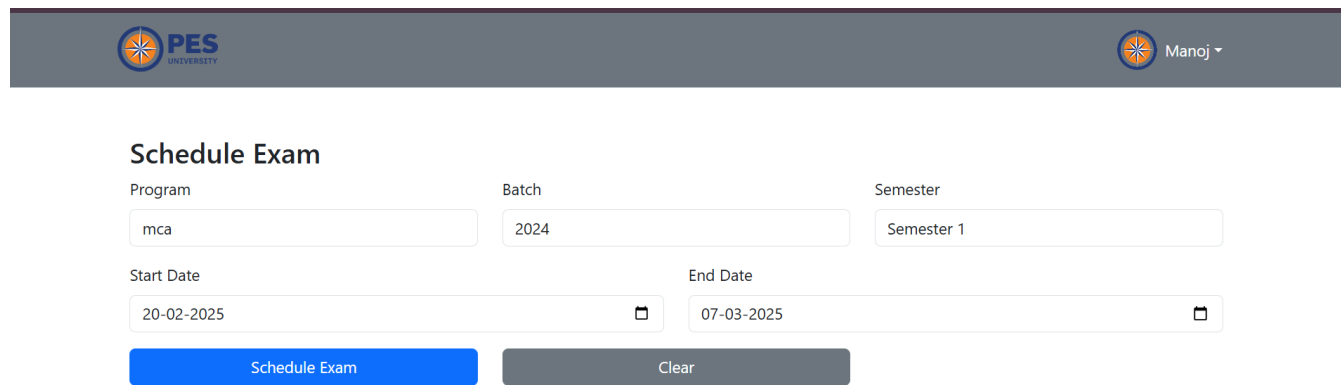


Figure 7.2: Admin Dashboard

The Exam Duty Allocator system's main control interface, the Admin Dashboard, provides a thorough rundown of the test scheduling procedure. Administrators can effectively allocate and oversee faculty responsibilities for different test sessions using this interface, guaranteeing a just and equitable division of labor. Additionally, the dashboard enables them to send users critical warnings or reminders, approve or reject faculty members' requests for duty swaps, and see and produce comprehensive information on faculty availability and duty allocations. The dashboard guarantees efficient and well-organized exam management by combining all essential administrative tasks into a single platform, improving transparency and streamlining processes.



Schedule Exam

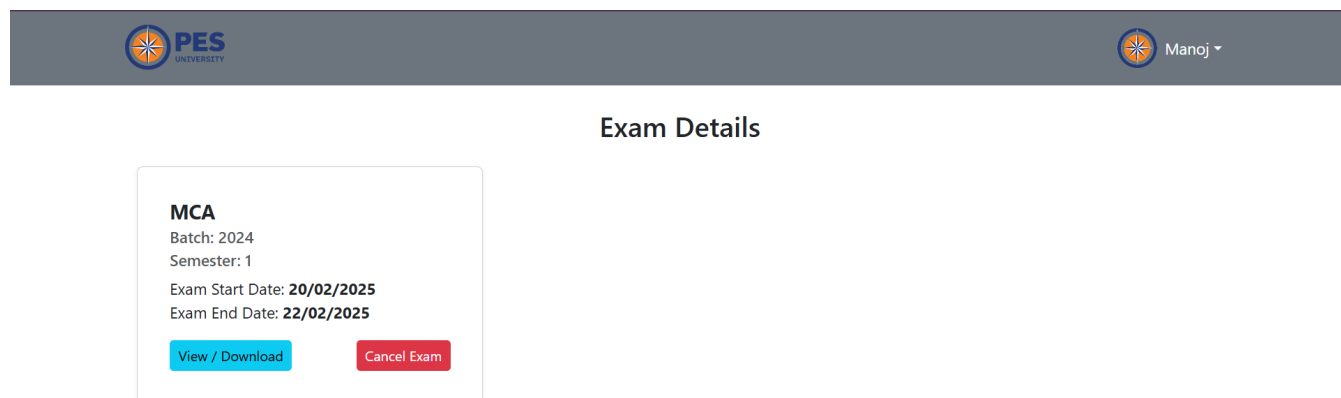
Program: mca Batch: 2024 Semester: Semester 1

Start Date: 20-02-2025 End Date: 07-03-2025

[Schedule Exam](#) [Clear](#)

Figure 7.3: Exam Schedule

Administrators can enter crucial information like the date, program, semester, start date, and end date on the Exam Schedule Input Page, guaranteeing a systematic and effective exam scheduling procedure.



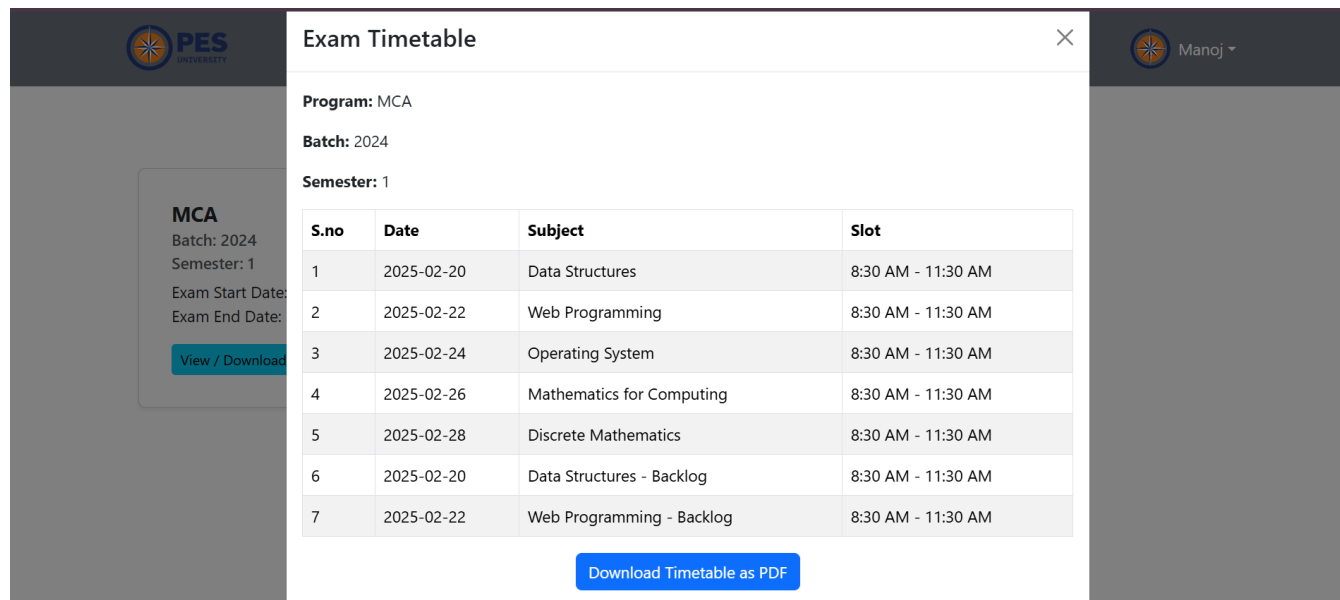
Exam Details

MCA
Batch: 2024
Semester: 1
Exam Start Date: **20/02/2025**
Exam End Date: **22/02/2025**

[View / Download](#) [Cancel Exam](#)

Figure 7.4: Manage Exam Schedules

For improved accessibility, administrators can view and download the exam schedule via the Manage Exam Schedule Page. It also offers the ability to postpone planned tests, guaranteeing flexibility in effectively managing exam schedules.



Exam Timetable

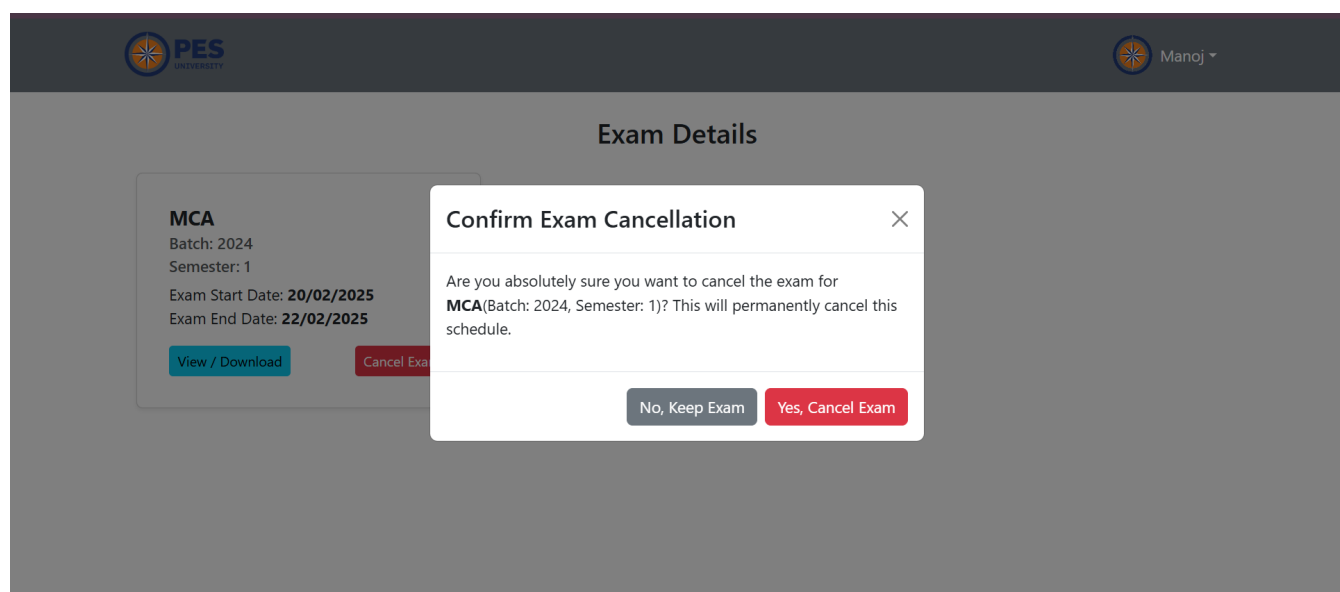
Program: MCA
Batch: 2024
Semester: 1

S.no	Date	Subject	Slot
1	2025-02-20	Data Structures	8:30 AM - 11:30 AM
2	2025-02-22	Web Programming	8:30 AM - 11:30 AM
3	2025-02-24	Operating System	8:30 AM - 11:30 AM
4	2025-02-26	Mathematics for Computing	8:30 AM - 11:30 AM
5	2025-02-28	Discrete Mathematics	8:30 AM - 11:30 AM
6	2025-02-20	Data Structures - Backlog	8:30 AM - 11:30 AM
7	2025-02-22	Web Programming - Backlog	8:30 AM - 11:30 AM

[Download Timetable as PDF](#)

Figure 7.5: View and Download

Users can download the timetable for offline usage and view the exam schedule in an organized style via the View and Download Page. For administrators and professors, this function guarantees accessibility and ease of reference.



Exam Details

MCA
 Batch: 2024
 Semester: 1
 Exam Start Date: **20/02/2025**
 Exam End Date: **22/02/2025**

[View / Download](#) [Cancel Exam](#)

Confirm Exam Cancellation

Are you absolutely sure you want to cancel the exam for **MCA**(Batch: 2024, Semester: 1)? This will permanently cancel this schedule.

[No, Keep Exam](#) [Yes, Cancel Exam](#)

Figure 7.6: Cancel Exam Schedule

Administrators can effectively remove or cancel planned exams using the Cancel Exam Schedule Page. It ensures appropriate management of exam schedules by offering the ability to choose a particular exam, check its information, and confirm the cancellation.

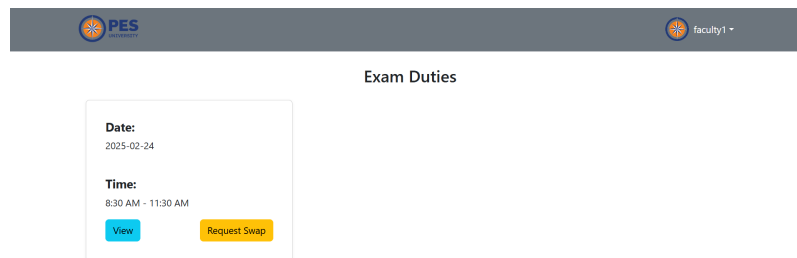


Figure 7.7: View Assigned Duties

Faculty members can view their assigned test obligations on the View Assigned obligations Page. It ensures openness and appropriate duty management by offering a clear schedule that shows the date and time.

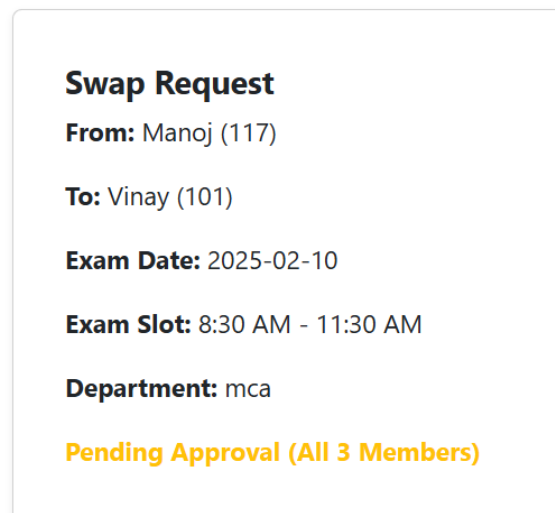


Figure 7.8: View Assigned Duties

Faculty members can check the status of their requested duty swaps using the swap request information tool. Three roles must approve the process: the administration, the department coordinator, and the desired faculty. This multi-level permission guarantees openness, responsibility, and efficient management of duty transfers.

Swap Request
From: Manoj (117)
To: Vinay (101)
Exam Date: 2025-02-10
Exam Slot: 8:30 AM - 11:30 AM
Pending Approval

Approve

Reject

Figure 7.9: View Assigned Duties

Shows how to approve or reject swap requests using the interface. Administrators and coordinators have access to all outstanding requests and can respond appropriately depending on their schedules and availability. This component maintains the balance of responsibility among faculty members by guaranteeing a clear and controlled approval process.

PESU - EXAM DUTY ALLOCATOR

Chapter 8: Testing

Chapter 8

Testing

8.1 Manual Testing

Test Case : Login Verification (Admin/Faulty)

Verify if admin and faculty users can successfully log in with valid credentials and receive error messages with invalid inputs.

MANUAL TESTING					
MODULE NAME: Authentication					
TEST ID	TC-1				
TEST NAME	Login Functionality				
TEST DESCRIPTION	Check if login with valid/invalid credentials behaves correctly				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Enter valid admin credentials	user: admin, pass: ****	Redirect to dashboard	Redirect to dashboard	Pass
2	Enter invalid credentials	user: wrong, pass: wrong	Show error message	Show error message	Pass
3	Leave credentials empty	-	Show validation error	Show validation error	Pass

Table 8.1: Test Case: Login Functionality

Test Case : Number of days for conducting exams

Validating the functionality that checks whether the number of available days is sufficient to automatically generate an exam timetable.

MANUAL TESTING					
MODULE NAME: Timetable Generation					
TEST ID	TC-2				
TEST NAME	Check Day Sufficiency for Timetable				
TEST DESCRIPTION	Ensure the system correctly validates the number of available days against number of exams and sessions required				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Input total number of subjects	12 subjects	System accepts subject input	System accepts subject input	Pass
2	Input available number of days	3 days	System accepts days input	System accepts days input	Pass
3	Input sessions per day	3 sessions/day	System accepts sessions input	System accepts sessions input	Pass

Table 8.2: Test Case : Check Day Sufficiency for Timetable Generation

Test Case : Rooms availability

Validating that the system checks if the number of available rooms is sufficient to conduct all exams scheduled per session.

MANUAL TESTING					
MODULE NAME: Room Allocation					
TEST ID	TC-3				
TEST NAME	Check Room Sufficiency				
TEST DESCRIPTION	Ensure that the system verifies room count based on total exams per session				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Input number of parallel exams in a session	10 subjects scheduled per session	System accepts exam schedule input	System accepts exam schedule input	Pass
2	Input available number of rooms	6 rooms	System accepts room count	System accepts room count	Pass
3	Trigger timetable generation	Click 'Generate Timetable'	System checks if 6 rooms ; 10 subjects	System shows "Insufficient Rooms" warning	Pass
4	Update room count and retry	10 rooms	System has enough rooms	System generates timetable successfully	Pass

Table 8.3: Test Case : Check Room Sufficiency for Exam Schedule

Test Case : Backlog Student Conflict

Validating that the system identifies conflicts when backlog students have overlapping exams in the generated timetable.

MANUAL TESTING					
MODULE NAME: Timetable Generation					
TEST ID	TC-4				
TEST NAME	Backlog Conflict Detection				
TEST DESCRIPTION	Ensure system checks for and prevents scheduling conflicts for backlog students				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Input regular and backlog exam data	Regular: 5 subjects, Backlog: 2 subjects	System accepts subject data	System accepts subject data	Pass
2	Assign backlog subjects to relevant students	Student A has 1 backlog	System maps backlog correctly	System maps backlog correctly	Pass
3	Trigger timetable generation	Click 'Generate Timetable'	System checks for student-wise subject clash	Conflict detected and alert shown	Pass
4	Adjust schedule to resolve clash	Move one exam to a different slot	Conflict resolved	Timetable generated without conflict	Pass

Table 8.4: Test Case : Backlog Conflict Detection in Exam Schedule

Test Case : Faculty Availability

Validating that the system checks if enough faculty members are available to invigilate all scheduled exams per session.

MANUAL TESTING					
MODULE NAME: Faculty Duty Allocation					
TEST ID	TC-5				
TEST NAME	Check Faculty Sufficiency				
TEST DESCRIPTION	Ensure system verifies faculty availability for invigilation before generating timetable				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Input number of exams per session	10 exams per session	System accepts exam count	System accepts exam count	Pass
2	Input available faculty count	6 faculties	System accepts faculty count	System accepts faculty count	Pass
3	Trigger timetable generation	Click 'Generate Timetable'	System checks if 6 faculties ; 10 exams	System shows "Insufficient Faculty" warning	Pass
4	Update faculty count and retry	12 faculties	System has enough faculties	Timetable generated successfully	Pass

Table 8.5: Test Case : Faculty Sufficiency Check in Exam Schedule

Test Case : Duty Swap Between Faculties

Validating that the system allows and properly processes exam duty swaps between two faculty members.

MANUAL TESTING					
MODULE NAME: Duty Swap Portal					
TEST ID	TC-6				
TEST NAME	Swap Duties Between Faculties				
TEST DESCRIPTION	Ensure that faculty A can swap duty with faculty B after mutual consent and admin approval				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Faculty A requests a swap with Faculty B	Faculty A: Slot 1, B: Slot 3	Request initiated and pending approval	Request initiated and pending approval	Pass
2	Faculty B confirms the swap	Accepts request in portal	Swap request moves to admin approval	Swap request moves to admin approval	Pass
3	Admin approves the swap	Click 'Approve'	Duties of A and B are interchanged	Duties updated successfully	Pass
4	Verify updated schedule	View updated schedule	Faculty A assigned Slot 3, B assigned Slot 1	Schedule updated as expected	Pass

Table 8.6: Test Case : Faculty Duty Swap

Test Case : Manual Duty Adjustment

Validating that the admin can manually modify the auto-assigned duties when necessary.

MANUAL TESTING					
MODULE NAME: Duty Allocation					
TEST ID	TC-7				
TEST NAME	Manual Override of Assigned Duty				
TEST DESCRIPTION	Ensure admin can manually reassign duties in the timetable after generation				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Open generated timetable	Auto-assigned duties shown	Timetable loads correctly	Timetable loads correctly	Pass
2	Select a faculty slot to change	Faculty X assigned Slot 2	Slot becomes editable	Slot becomes editable	Pass
3	Assign a different faculty	Replace Faculty X with Faculty Y	Assignment saved	Assignment saved	Pass
4	Verify updated assignment	View updated schedule	Faculty Y now shown in Slot 2	Updated correctly	Pass

Table 8.7: Test Case : Manual Adjustment of Duties by Admin

Test Case : Download Generated Timetable

Validating that the system allows the admin to download the generated exam timetable in a readable format.

MANUAL TESTING					
MODULE NAME: Timetable Management					
TEST ID	TC-8				
TEST NAME	Download Generated Timetable				
TEST DESCRIPTION	Ensure the system allows the admin to download the timetable in a readable format (e.g., PDF, Excel)				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Generate the timetable	Click 'Generate Timetable'	Timetable generated successfully	Timetable generated successfully	Pass
2	Click 'Download' button	Click 'Download PDF' button	File download begins	File download begins	Pass
3	Verify file format	Downloaded file	File is in the correct format (PDF/Excel)	File in correct format	Pass
4	Open downloaded file	Open downloaded timetable	Timetable opens correctly with no corruption	Timetable opens correctly	Pass

Table 8.8: Test Case : Download Generated Timetable

Test Case : Download Exam Duties Details in Excel Format

Validating that the system allows the admin to download the detailed exam duties in an Excel format.

MANUAL TESTING					
MODULE NAME: Duty Report Management					
TEST ID	TC-9				
TEST NAME	Download Exam Duties in Excel Format				
TEST DESCRIPTION	Ensure the system allows the admin to download detailed exam duties in Excel format				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Open duty report page	Click 'Exam Duty Report'	Duty report page loads with details	Report page loads correctly	Pass
2	Click 'Download Excel' button	Click 'Download Excel' button on the page	Excel file download begins	File download begins	Pass
3	Verify downloaded file format	Open downloaded file	File is in Excel format (.xlsx) and contains exam duty details	File in correct format with all details	Pass
4	Verify file content	Open Excel file	Exam duties, faculties, and timeslots are displayed correctly in the file	All data correctly displayed in columns	Pass

Table 8.9: Test Case : Download Exam Duties in Excel Format

Test Case : Cancel Entire Exam and Associated Duties

Validating that the system allows the admin to cancel an entire exam, which will automatically cancel all duties associated with the exam.

MANUAL TESTING					
MODULE NAME: Exam Cancellation					
TEST ID	TC-10				
TEST NAME	Cancel Entire Exam and Duties				
TEST DESCRIPTION	Ensure the admin can cancel an entire exam, which will automatically cancel all associated duties				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Open exam schedule page	View the list of scheduled exams	All exams are listed	Exams listed correctly	Pass
2	Select the exam to cancel	Click 'Cancel Exam' button next to the selected exam	System asks for confirmation to cancel the exam	System prompts confirmation message	Pass
3	Confirm cancellation	Click 'Yes' to confirm cancellation	Exam is canceled and all associated duties are removed from the timetable	Exam and duties canceled	Pass
4	Verify cancellation in the system	Check the exam schedule and duty allocations	The canceled exam and its duties should no longer be listed in the system	Exam and duties removed	Pass

Table 8.10: Test Case : Cancel Entire Exam and Associated Duties

Test Case : Faculty View and Download Duty

Validating that the faculty can view their assigned duties and download their duty details in a readable format.

MANUAL TESTING					
MODULE NAME: Faculty Duty Management					
TEST ID	TC-11				
TEST NAME	Faculty View and Download Duty				
TEST DESCRIPTION	Ensure that the faculty can view their assigned duties and download their duty details in a readable format (e.g., PDF, Excel)				
STEP#	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Login as faculty	Enter faculty credentials	Faculty dashboard loads with assigned duties	Dashboard loads correctly with duties	Pass
2	View assigned duties	Click on 'View Duties' option	Assigned duties for the faculty are displayed with relevant details	Assigned duties displayed correctly	Pass
3	Download duty details	Click 'Download Duty Details' button	Duty details are downloaded in the selected format (PDF/Excel)	Duty details downloaded in the selected format	Pass
4	Verify downloaded file	Open downloaded file	File is in the correct format (PDF/Excel) and contains the faculty's duties correctly displayed	File in correct format with all duties	Pass

Table 8.11: Test Case : Faculty View and Download Duty

PESU - EXAM DUTY ALLOCATOR

Chapter 9: Conclusion

Chapter 9

Conclusion

The difficulties of allocating, overseeing, and monitoring exam responsibilities for faculty members at educational institutions are effectively addressed by the Exam Duty Allocator project. The entire procedure is automated by the system, which guarantees a fair division of duties, lessens administrative burden, and offers a simplified method for the faculty and the Controller of Examinations (COE).

Important features like automated duty allocation, manual overrides, faculty and room administration, timetable production, and duty swap capability were added as the system was being developed. A user-friendly interface and dependable back-end processing were guaranteed by the usage of web technologies such as React.js, Node.js, and MongoDB. Furthermore, by giving faculty members real-time access to their calendars, the technology promotes transparency and flexibility in the management of responsibilities.

The system is resilient and scalable due to its capacity to handle a variety of edge circumstances, including competing exam schedules, instructor availability, and room sufficiency. Furthermore, a key component that simplifies test-related procedures is the ability for faculty to download their duty schedules and for the COE to create and modify exam timetables.

PESU - EXAM DUTY ALLOCATOR

Chapter 10: Future Work

Chapter 10

Future Work

Although the Exam Duty Allocator project has effectively met its primary goals, there are still a number of areas that could use improvement. The incorporation of AI-based task allocation is one possible improvement; it would minimize manual intervention by automatically recommending the most effective duty distribution based on variables including workload distribution, faculty preferences, and historical performance. The system might also be expanded to a mobile application, which would increase accessibility and convenience by enabling teachers and COE employees to view duty schedules and receive notifications while on the road. In order to ensure smooth synchronization of exam schedules and task assignments, future revisions of the project should additionally concentrate on connecting with currently used exam scheduling software.

Real-time conflict identification, which would take into account more complicated situations like exam paper overlaps, instructor unavailability, and student-specific requirements, is another crucial aspect for future development. A reporting and analytics feature could help the COE make data-driven decisions by offering insightful information about faculty workloads, duty allocation trends, and overall exam schedule. To guarantee scalability, high availability, and ease of maintenance, the system may also be deployed to the cloud, making it usable by organizations with several campuses.

PESU - EXAM DUTY ALLOCATOR

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Appendix A: Bibliography

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PESU - EXAM DUTY ALLOCATOR

Appendix B: User Manual

Appendix B: User Manual

Admin Role

This section outlines the operational flow and features available to the Admin (COE/Exam Cell) in the **PESU – Exam Duty Allocator** system.

1. Auto Timetable Scheduling with Duty Allocation

- Navigate to the **Schedule Timetable** section.
- Provide inputs:
 - Program
 - Batch
 - Semester
 - Start Date
 - End Date
- Click on **Generate Timetable** to create the exam schedule and auto-allocate duties.

2. Manage Timetable

- Go to the **Manage Timetable** section.
- View and verify scheduled exams.
- Use **Cancel Timetable** to delete a schedule if needed.
- Click **Download as PDF** to export the exam timetable in a printable format.

3. Approve or Reject Duty Swap Requests

- Navigate to the **Duty Swap Requests** panel.
- View pending swap requests submitted by faculty.
- Approve or reject requests based on feasibility and availability.

4. Manual Duty Adjustment

- Open the **Manual Adjustment** section.
- Select the appropriate timetable and exam.
- Choose the currently assigned faculty member.
- Select a replacement faculty from the available list.
- Click **Adjust Duty** to confirm the changes.

5. Download Duty Allocation as Excel

- Go to the **Reports or Duty Allocation** section.
- Click **Download as Excel** to export the duty assignment list.

6. Sample Workflow

- Schedule exams for a given semester with date range and batch details.
- Review or edit using Manage Timetable.
- Approve any swap requests or make manual adjustments.
- Download the final duty allocation in Excel format for record-keeping or circulation.

Department Coordinator Role

This section outlines the responsibilities and features available to the Department Coordinator in the **PESU – Exam Duty Allocator** system.

1. Approve or Reject Duty Swap Requests

- Navigate to the **Duty Swap Requests** section.
- Review the list of swap requests raised by faculty within your department.
- Each request has two options:
 - **Approve** – Confirms the duty swap.
 - **Reject** – Declines the swap request based on departmental feasibility.
- Decisions at this level apply only to requests within the Department Coordinator's department.

2. Download Department Duty Allocation as Excel

- Go to the **Reports or Duty Allocation** section.
- Filter or select your department.
- Click **Download as Excel** to export the duty allocation list for your department.

Faculty Role

This section describes how faculty members can interact with the PESU – Exam Duty Allocator portal to manage and track their exam duties.

1. View Exam Duties

- Navigate to the **Exam Duties** section.
- View the list of all assigned exam duties with **date, time, and venue**.
- Click the **Download as PDF** button to save a copy of your duty schedule.
- For each duty, a **Swap Request** button is available.
- To request a swap:
 - Click on the **Swap Request** button for a specific duty.
 - Select another faculty member to swap with from the available list.
 - Submit the request for further approvals.

2. Request Swap

- Go to the **Requested Swaps** section to track the status of your swap requests.
- Each entry will show:
 - The duty details (exam, date, time)
 - Status of the request:
 - * Approved/Rejected by the other faculty
 - * Approved/Rejected by the Department Coordinator
 - * Approved/Rejected by the Admin

3. Approve Swap

- Navigate to the **Approve Swaps** section.
- Here, faculty can view swap requests sent by others.
- Each request has two options:
 - **Accept** – Agree to take up the duty.
 - **Deny** – Decline the swap request.

PESU - EXAM DUTY ALLOCATOR

Appendix C: Plagiarism and AI Report

Appendix C: Plagiarism and AI Report

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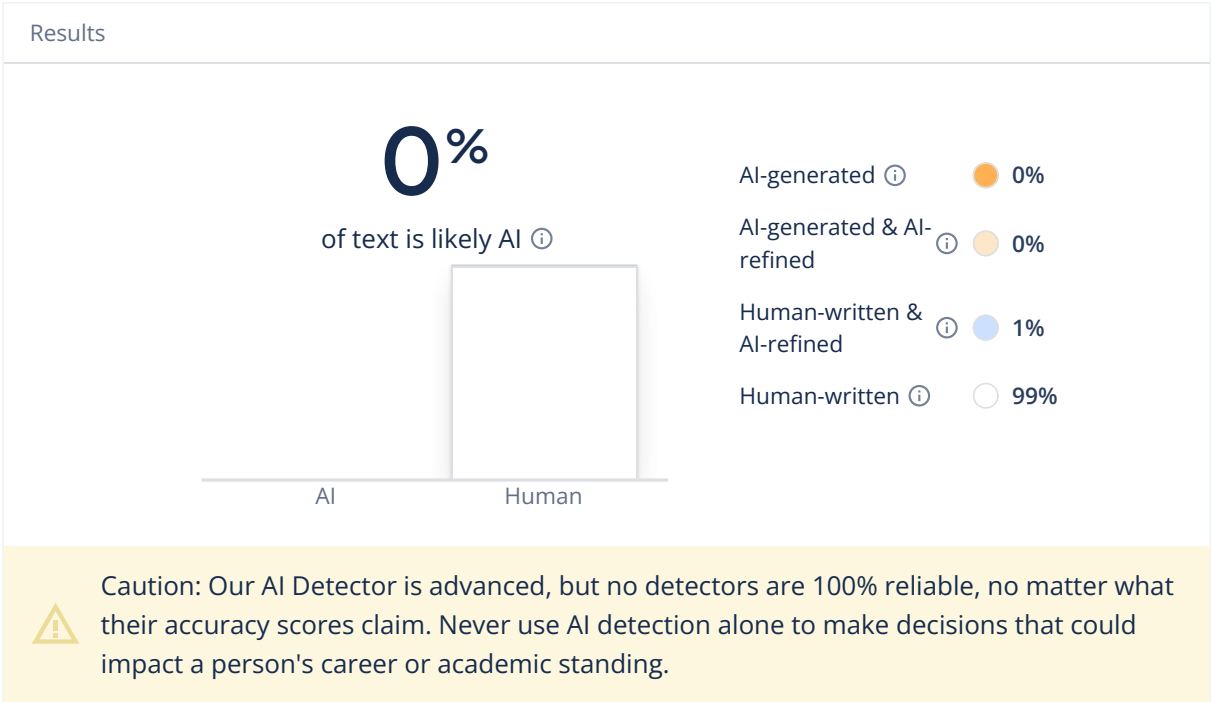
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PESU - EXAM DUTY ALLOCATOR

Appendix D: Poster

Appendix D: Poster



PES University

Department of Computer Applications

PESU Exam Duty Allocator

What if exam duties could be allocated in seconds—fairly, efficiently, and without conflicts?

1. Abstract

PESU - EDA automates exam duty scheduling to ensure fairness, reduce conflicts, and ease administrative workload.

2. Challenge

Manual duty allocation is time-consuming, error-prone, and lacks transparency.

3. Tools




mongoDB

4. Results



5. Conclusion

The system enhances efficiency, ensures equitable workload distribution, and minimizes manual intervention.



Developed by,
Manoj S
PESIPG23CA078

Under the guidance of,
Mr. Dilip Kumar Maripuri,
Associate Professor

Exam Timetable

Program: MCA
Batch: 2023
Semester: 4

S.no	Date	Subject	Slot
1	2023-02-01	Cloud Computing	8:30 AM - 11:30 AM
2	2023-02-03	Cybersecurity	8:30 AM - 11:30 AM
3	2023-02-05	Big Data Analytics	8:30 AM - 11:30 AM
4	2023-02-07	Mobile Application Development	8:30 AM - 11:30 AM
5	2023-02-10	Advanced Databases	8:30 AM - 11:30 AM

[Download Timetable as PDF](#)

Adjust Duty - Advanced Databases
Currently Assigned Faculty

Name: Ravi
Email: Ravi@gmail.com
Employee ID: 102

[Adjust Duty](#)

Name: Priya
Email: Priya@gmail.com
Employee ID: 105

[Adjust Duty](#)

Name: Meera
Email: Meera@gmail.com
Employee ID: 107

[Adjust Duty](#)